

Luck in a Puzzling World: An Essay

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Abstract

Luck is easy to invoke after an outcome and difficult to identify before it. This essay asks what realized success and failure can legitimately tell us about strategy, capability, and merit. It separates what people choose from what could happen, what actually happens, and the outcomes that follow. This distinction helps explain why rarity, randomness, and unexplained variation do not by themselves identify luck, and why successful experience does not automatically become generalizable expertise. The analysis then follows success beyond the initial outcome: prior success can reshape future opportunities, while explanations of success can spread through imitation and social learning. The resulting perspective calls for greater discipline in attributing outcomes to merit, strategy, or luck, with implications for how we interpret success, expertise, and persistent advantage.

Keywords: luck; uncertainty; realization; success; attribution; transportability; cumulative advantage

1. Why Luck Bothers Us

People choose actions, but they do not choose outcomes.

The distinction sounds obvious until we try to explain success. A founder builds a company and becomes enormously wealthy. An investor follows a disciplined strategy for decades and accumulates a fortune. A scientist makes a discovery that changes a field. A young employee joins an obscure company shortly before it becomes an industry leader. Once we know how these stories end, the path leading to the outcome tends to become cleaner. The successful strategy becomes visible. The failed alternatives fade. Decisions that were made under uncertainty begin to look like steps toward a destination.

That is one reason luck is such an intriguing subject. We use the word easily, yet it becomes difficult to say exactly what we mean by it.

Consider a lottery winner. Few people object to saying that someone who buys a Powerball ticket and wins the jackpot was lucky. The person's action was necessary—without the ticket there is no prize—but buying the ticket plainly does not explain why that particular ticket won. The distinction between action and realization seems easy.

Now replace the lottery ticket with an investment strategy. Warren Buffett's extraordinary record plainly cannot be understood as a lottery drawing. Skill, judgment, discipline, institutional arrangements, capital, learning, and repeated decisions matter. But another question remains: how much does the extraordinary outcome itself tell us about how extraordinary each of those inputs must have been?

That question is harder than it first appears. If we begin with the winner and work backward, almost every feature of the winner can acquire explanatory significance. Patience becomes the reason for success. Concentration becomes the reason. Independence of thought becomes the reason. The ability to wait

becomes the reason. Each may genuinely matter. Yet the fact that a characteristic can be found in a successful person's history does not establish how much of the success it caused.

The problem is symmetric. A failed entrepreneur may have made poor decisions. Or a sensible strategy may have encountered an unfavorable realization. A scientist whose promising project produced nothing may have lacked insight, or the relevant evidence may simply not have appeared. A good decision can end badly; a bad decision can end well.

Experimental research shows how difficult it is for people to preserve this distinction after learning the result. Baron and Hershey found that people evaluated decisions and decision-makers more favorably when outcomes were favorable, even when the information available to the decision-maker was held constant [1]. The outcome reaches backward and changes our interpretation of the choice.

Selection makes the problem worse. We study winners because they won. Denrell showed how learning from surviving and successful organizations can generate false beliefs about effective management practices when failures are undersampled [2]. The missing cases are not merely an inconvenience. They can change the apparent relationship between a practice and success.

This does not tell us that winners are ordinary, that skill is unimportant, or that success is mostly luck. It tells us something more basic: **an extraordinary outcome is not a transparent record of how it was produced.**

That puzzle reaches well beyond famous investors and entrepreneurs. It appears whenever outcomes become evidence about merit. We promote managers whose projects succeeded. We allocate capital to investment managers with strong records. We imitate successful organizations. We ask accomplished people for advice. We assign credit for success and blame for failure. In each case, the observed outcome becomes information about the person and the strategy.

Sometimes that inference is justified. Sometimes it is not. The difficult part is knowing which case we are in.

This is where a study of luck becomes more than an exercise in defining a familiar word. The subject sits at the boundary between what people do and what happens to them, between agency and uncertainty, and between a realized history and the alternatives that disappeared when that history became real.

This essay asks why luck, merit, strategy, and outcomes become so difficult to separate once the outcome is known.

2. What Do We Usually Mean by Luck?

Imagine four situations.

A person buys a lottery ticket and wins a jackpot.

A researcher notices an unexpected result, recognizes its importance, and turns it into a discovery.

A commuter takes a different route one morning and happens to meet the person who later offers a career-changing opportunity.

A fund manager produces an unusually strong return, but statistical analysis suggests that results this good can sometimes occur by chance even without exceptional investment skill.

All four situations may be described using the word *luck*. They are not describing the same thing.

In the lottery, luck seems closely related to chance. The mechanism is intentionally probabilistic. The winner does not control the drawing.

In the scientific case, "luck" is less satisfactory. An unexpected observation may occur without being sought, but recognizing its value can require preparation, expertise, curiosity, and action. Research on serendipity has therefore distinguished it from generic luck. Recent synthesis work emphasizes surprise, agency, and value as central elements of serendipity [3]. The unexpected event matters, but so does what the person does with it.

The commuter's encounter sounds like "right place, right time." Career research has studied such chance events directly. Studies report that people frequently perceive unplanned encounters and events as influencing career decisions and transitions, and some work explicitly identifies being in the right place at the right time as an influential positive career event [4–7]. Here luck seems connected to timing and position rather than to a formal randomizing device.

The fund manager introduces yet another meaning. In empirical finance, "luck" may be defined relative to a benchmark model: how much apparent performance could arise from sampling variation if true abnormal skill were absent? Bootstrap studies construct distributions of performance under such null hypotheses and compare observed managers with those distributions [8]. This is a useful technical meaning of luck, but it is conditional on the benchmark. Change the model and the inference can change.

Philosophy adds further distinctions. Accounts of luck have appealed to low probability, lack of control, significance to the affected person, and what philosophers sometimes call modal fragility—the idea that a lucky event could easily have gone otherwise in nearby possible circumstances [9,10]. The literature on moral luck asks a related but different question: how should responsibility be understood when outcomes depend partly on circumstances beyond an agent's control? [10]

These interpretations overlap, but none simply contains all the others.

Low probability, for example, seems promising until we shuffle a deck of cards. Every exact ordering of a properly shuffled deck is extraordinarily improbable. Yet we do not normally exclaim that the resulting sequence was extraordinarily lucky. Something more is needed—perhaps significance, prior specification, control, or consequence.

Lack of control is also too broad. Tomorrow's sunrise is outside an individual's control, but observing it is not ordinarily considered lucky. Conversely, a person may deliberately increase the probability of encountering favorable opportunities without controlling which opportunity actually appears. Career theories of planned happenstance make exactly this distinction: exploratory behavior can increase exposure to unplanned opportunities without turning those opportunities into predetermined outcomes [11].

"Right place, right time" has intuitive appeal, but it raises another question. Is that simply a convenient retrospective description? Once a career succeeds, it is easy to identify the meeting, location, institution, or historical moment that now appears decisive. How would we have identified the same event before knowing where the trajectory ended?

Even randomness is not enough. A random event can be trivial. A deterministic event can be experienced as fortunate if the person could not reasonably know or control it. And an event that is fortunate for one person may be unfortunate for another.

The familiar word *luck* therefore appears to cover several different relationships:

chance and realization; probability and surprise; control and responsibility; opportunity and agency; timing and position; statistical expectation and observed performance; event and personal significance.

That diversity may eventually imply that there is no single thing called luck. Or it may mean that these interpretations share a deeper structure we have not yet identified. At this stage, the more useful move is not to choose a definition but to keep the distinctions visible.

3. Some Easy Answers That Are Not So Easy

A more consequential analytical shortcut is to define luck as whatever a model cannot explain.

Suppose:

$$Y=f(X)+\epsilon.$$

If X includes intelligence, effort, education, capital, strategy, experience, and other observable factors, it is tempting to call the residual ϵ "luck."

The residual tells us how much the model leaves unexplained; it does not tell us what the unexplained portion consists of.

Residuals can contain measurement error, omitted variables, misspecification, unstable relationships, sampling variation, and unobserved interactions. If a model leaves Warren Buffett's wealth largely unexplained, the residual tells us that the model did not explain the wealth. It does not tell us what produced the unexplained portion.

Empirical studies of investment performance make this point concrete. Researchers can ask whether unusually high returns are consistent with chance under a specified asset-pricing model. But conclusions about skill and luck can depend on the benchmark, the bootstrap procedure, the treatment of dependence across funds, and whether skill is allowed to vary over time [8,12,13]. "Luck" in such a model is not a newly discovered force. It is an inference conditional on an analytical structure.

Another shortcut runs in the opposite direction: if an outcome is repeated, perhaps luck disappears.

Repeated success certainly provides more information than a single success. A person who performs extraordinarily once may have experienced a favorable realization. Someone who performs extraordinarily over many independent opportunities gives us stronger reason to infer persistent capability.

But repeated outcomes are not always independent. Success can change the conditions of the next round.

A successful investor gains more capital and access. A successful scientist gains reputation, collaborators, and resources. A successful founder gains networks and the ability to recruit. A celebrated author gets a larger audience for the next book. The next opportunity set is no longer the one that existed before the first success.

The sociology of science has long described such cumulative advantage through the "Matthew effect" [14]. Randomized field experiments have gone further by bestowing arbitrary initial advantages in live social systems and observing whether those advantages generated additional success. Across several settings, they did [15].

Repeated success can therefore contain at least two kinds of information at once. It can reveal persistent capability, and it can reflect feedback from earlier success.

A final shortcut is to reverse the argument: if outcomes cannot be cleanly attributed to skill, they must be luck.

The reverse inference has the same weakness.

A poor explanation of merit is not positive evidence for luck. A gap in knowledge is a gap in knowledge.

This point becomes important because research on luck can easily become self-sealing. If success is unexplained, call it luck. If the outcome is rare, call it luck. If two apparently similar people diverge, call the difference luck. Under those rules, every failure of the model becomes confirmation of the concept.

A useful study of luck must demand more.

4. What Happened Between the Choice and the Outcome?

Consider a simple investment decision.

At the end of today, an investor chooses a strategy. Tomorrow several things might happen. Markets may rise or fall. A company may announce news. Another investor may offer a transaction. Financing may remain available or disappear. Nothing important may happen at all.

The investor chooses the strategy today. The investor does not choose which tomorrow arrives.

That observation suggests a simple way to slow down the story:

$$S(t-1) \rightarrow \mathcal{X}(t) \rightarrow X(t) \rightarrow Y(t)$$

The notation is less important than the separation it forces.

$S(t-1)$ is the strategy or action selected before the next period is realized.

$\mathcal{X}(t)$ represents the possible states that might follow. Strategy can change this set and can change the probabilities within it. Buying a lottery ticket creates a possibility of winning that did not exist without the ticket. Developing expertise can create opportunities that would not be available to an unprepared person. Moving to a particular city can change whom someone might meet. Holding cash during a market crisis can change which investments are feasible.

Then one state, $X(t)$, actually realizes.

Finally, that realized state contributes to the outcome $Y(t)$.

The distinction matters because ordinary explanations often jump directly from:

$$S(t-1)$$

to:

$$Y(t).$$

If the result is good, the strategy looks good. If the result is extraordinary, the strategy may begin to look extraordinary.

But strategy and realization perform different analytical jobs.

A sensible strategy may improve:

$$P(X_{\text{good}}(t) | S(t-1))$$

without making:

$$P(X_{\text{good}}(t) | S(t-1)) = 1$$

This gives us a way to preserve both agency and uncertainty.

It also tells us something positive. Evidence that a strategy repeatedly improves the distribution of feasible outcomes is evidence about strategy quality, even when no single realized outcome proves that quality by itself.

A person can deserve credit for choosing a strategy that improves the distribution of possible outcomes without deserving causal credit for every favorable state that subsequently occurs. Conversely, a person can choose a poor strategy and still receive a favorable realization.

Career research provides an intuitive parallel. Planned-happenstance theory does not tell people to wait passively for luck. It emphasizes exploratory behavior that can increase exposure to unexpected opportunities [11]. Later research on career chance events similarly finds that unplanned events can be influential while actors still exercise agency in recognizing and responding to them [4–7].

Scientific serendipity follows a similar pattern. The unexpected observation may not be chosen, but recognizing its importance and converting it into knowledge can require substantial capability [3].

The distinction also helps with a thought experiment about tomorrow's stock price.

Imagine that you somehow know tomorrow's closing price today. The information could be enormously valuable because you possess it while an action remains possible. Now imagine receiving exactly the same information one minute after tomorrow's market closes. The informational content is identical. Its practical value is not.

The difference is temporal position.

This tells us something about $X(t)$ that a static list of factors can obscure. States are not merely things an actor possesses. They occur at particular times, in particular positions, relative to particular opportunities.

That insight will matter later. For now, the important point is simpler.

When we observe:

$$Y(t),$$

we are looking at the end of a process.

The outcome does not automatically tell us which parts were created by strategy, which were realized from among several possibilities, which depended on timing or position, and which were amplified after the initial event.

This is the missing middle in many stories of success and failure.

And it may be the most useful place to begin looking for luck—not by assuming that luck occupies the middle, but by asking what happened there.

5. The World Does Not Hold Still

The missing middle also moves through history. Imagine two investors who follow the same sensible rule: keep cash available and buy strong businesses when fear has pushed prices far below reasonable estimates of value.

The missing middle is therefore historically situated. Move the same strategy through time and it may no longer be the same practical strategy.

This is why time is awkward to treat as merely another item in a list of explanatory factors. A person's resources, knowledge, relationships, and strategy exist at particular moments. Their consequences depend partly on what else is possible at those moments. A compact reminder is:

$$Y(t) = f_t(X(t))$$

Career research encounters the same problem in ordinary language. Studies of chance events find that careers can be altered by unplanned meetings, project assignments, institutional changes, and being in the "right place at the right time" [6,7,16]. Planned-happenstance theory goes further: people can deliberately explore and increase exposure to unexpected opportunities without determining which opportunity will appear [11].

Time locates an actor historically; situated position locates the actor within the surrounding social, institutional, and informational environment.

Time is only one part of the problem. Two similarly trained professionals may occupy very different positions in organizations, networks, jurisdictions, markets, or flows of information. One is assigned to a project led by a future sponsor; another joins a project that is cancelled. Calling this merely "spatial" is too narrow. The broader idea is **situated position**.

This creates a transportability problem. If a successful executive says, "This is what worked for me," what must be true before "for me" becomes "for you"? Causal-inference research formalizes versions of this question across populations and environments [17]. The original success cannot answer it by itself.

An old saying emphasizes favorable timing, advantageous circumstances, and harmony among people. It captures a durable intuition: what someone can accomplish depends partly on when action occurs, where the person is situated, and how the person is related to others.

The research problem is to determine when those differences actually matter—and when they are stories we discover only after knowing the outcome. That question immediately encounters a more fundamental difficulty: we never observe everything that could have mattered.

6. The Things We Cannot See

Imagine two sibling dancers performing the same choreography on the same stage at the same time.

One hand rises slightly higher. One turn begins a fraction earlier. Their performances are not identical, and a director may notice one rather than the other.

The example creates a problem for any attempt to study luck by finding two "identical" people. There are no identical people, and no empirical record contains every causally relevant feature of two lives.

Suppose:

$$X_A(\text{obs}) \approx X_B(\text{obs})$$

but later:

$$Y_A \gg Y_B$$

We have not thereby identified luck. We may have measured something badly, omitted a relationship or experience, missed an interaction, or used the wrong model. The strongest conclusion may be only that our measured variables do not fully explain the divergence.

Historical cases impose an additional ceiling. We cannot rerun a person's life while holding everything fixed and changing one realization. The counterfactual investor who made the same decisions but encountered different markets is unavailable.

Extreme cases make this worse. If we select someone because the outcome is extraordinary, we have conditioned the investigation on the thing we later hope to explain. Regression toward the mean, survivor selection, and winner-selection effects all warn against reading an extreme outcome as transparent evidence of an extreme underlying trait; inference after selecting apparent winners formalizes this problem directly [21].

The problem cuts both ways. A billionaire's wealth does not by itself establish extraordinary merit. Failure to explain that wealth with our variables does not establish extraordinary luck either.

A useful discipline follows:

Luck cannot be inferred from ignorance.

Positive evidence is possible. When a realization difference is independently observed, precedes the divergence, and is relevant to the outcome, it tells us something about the realized path. Complete causal decomposition is not required for that narrower inference.

Positive evidence should identify something that happened between the prior strategy and later outcome: a documented encounter, allocation, draw, timing difference, institutional change, or other realization not defined merely because the outcome became exceptional.

Sometimes such evidence exists. Often it does not. That limit may itself be one of the most important findings available to research on luck.

7. When Winners Explain Why They Won

Imagine two decision-makers facing the same uncertain problem from the same information. One receives a favorable outcome and the other an unfavorable one. Which made the better decision?

Experiments on outcome bias show how difficult it is to keep decision quality separate from result. Knowing that a decision ended well makes both the decision and the decision-maker look better [1].

Now replace the experimental subject with an entrepreneur. The company succeeds. Journalists ask what made the difference. The founder explains persistence, hiring, focus, customer obsession, speed, or independence of thought. None of those lessons is necessarily wrong. The problem is that the history was selected by the outcome.

Unsuccessful founders who were equally persistent do not receive the same audience. Denrell's work makes this selection problem formal: learning disproportionately from surviving organizations can make practices appear associated with success even when the relationship differs in the full population [2].

The social story compresses:

$$S(t-1) \rightarrow \mathcal{X}(t) \rightarrow X(t) \rightarrow Y(t)$$

into:

$$S \rightarrow Y.$$

Then another step follows:

$$Y^+ \rightarrow \text{perceived competence} \rightarrow \text{authority}$$

Successful people may possess deep tacit knowledge. The narrower question is what part of that knowledge is demonstrated by the outcome itself. A person observes one realized trajectory, not the distribution that would have followed the same strategy across other people, times, and circumstances.

Hence:

Outcome authority is not predictive authority.

The distinction matters because winners do not merely explain themselves. Other people listen, and organizations, investors, and policymakers may act on advice whose transportability has never been established.

8. When Success Starts Producing More Success

Suppose an early success was partly contingent. What happens next?

Many social systems do not reset after each round. A young scientist who receives an early award gains visibility and resources. A founder whose first product succeeds can raise capital more easily. A successful investor controls a larger capital base and may receive access unavailable to smaller investors.

The first outcome changes the next possibility set:

$$Y^+(t) \rightarrow \mathcal{X}_{\text{new}}(t+1)$$

In increasing-returns settings, even relatively small historical events can alter adoption paths and contribute to later lock-in, providing a formal example of how early realizations can reshape subsequent possibilities [22].

This separates the origin of an advantage from its persistence.

Merton's Matthew effect described how already eminent scientists can receive disproportionate recognition [14]. Later randomized field experiments bestowed arbitrary initial advantages in live social systems and found that these advantages increased subsequent rates of success across several settings [15].

That does not tell us that persistent winners lack superior ability. It tells us that prior success can itself become one cause of later success.

Repeated extraordinary performance may therefore contain several processes at once: persistent capability, learning, compounding, reputation, access, and feedback.

It helps to separate four questions: formation concerns the initial divergence; amplification its enlargement; persistence its durability; and transmission its movement across people, organizations, or generations.

Formation→Amplification→Persistence→Transmission.

What produced the initial divergence? What made it larger? What kept it from disappearing? What allowed it to pass across organizations, networks, or generations?

The answers need not be the same.

9. When Success Stories Change the World

The effects of a successful outcome do not stop with the successful person.

Other people watch. Once they observe the outcome, they begin forming explanations of what produced it.

An investment strategy performs exceptionally well, and capital flows toward similar strategies. A company succeeds with an unusual practice, and competitors imitate it. Consultants package it. Business schools teach it. What began as one history becomes a prescription.

The sequence extends:

$Y(t) \rightarrow \text{attribution} \rightarrow \text{social knowledge} \rightarrow \text{imitation} \rightarrow X(t+1)$

The interpretation of the past begins changing the future.

Social-learning research shows why this need not require individually foolish people. Observing others can be rational because their actions contain information. Yet under some conditions the same process produces informational cascades in which collective behavior becomes fragile or wrong [18].

Management-fashion research describes how practices can become collectively accepted and spread through consultants, gurus, media, and professional institutions [19]. Narrative economics examines how contagious stories can shape expectations and economic behavior [20].

The feedback can become especially puzzling when an incomplete lesson from success changes behavior in a way that temporarily makes the lesson appear correct. The belief begins producing evidence for itself.

The general sequence is:

outcome→story→behavior→new outcome.

A society can therefore do more than misunderstand a successful event. It can institutionalize an interpretation of it.

10. So, Does Luck Exist?

We can now return to the question we postponed.

A lottery winner seems an easy case. But what common mechanism connects a lottery drawing, an accidental meeting, a competitor's mistake, an unexpected scientific observation, and a timely market opportunity?

One established possibility is modal: an event is lucky when it could easily have gone otherwise. That captures an important intuition but raises difficult questions about which alternatives count as nearby [9].

Perhaps luck is relational: an event is lucky *for someone* because it matters to that person and because the realization was not determined by what that person knew, chose, or controlled beforehand.

Or perhaps luck is not one thing. The word may gather together stochastic realization, favorable circumstances, timing, positional advantage, chance encounters, unexpected opportunities, model-relative sampling variation, and ex-post attribution.

That possibility does not make the concept useless. It changes the research question from "What is the one mechanism of luck?" to "Which relationships among agency, realization, control, significance, and consequence are being described when we use the word?"

A modest candidate remains: luck may be worth investigating as something about the **relationship between an actor and a realized transition**, viewed from an ex-ante information and control boundary.

That is a direction, not a definition.

11. Why the Answer Might Matter

Suppose future research finds that repeated extraordinary outcomes remain strongly associated with persistent differences in capability and strategy after context and feedback are addressed. Merit-based interpretations would become more defensible, and learning from winners would have a stronger empirical foundation.

Suppose instead that comparable strategies frequently lead to materially different realizations. Outcomes would become noisier measures of merit. Credit for success—and blame for failure—would deserve more caution.

Suppose small initial contingencies are strongly amplified by capital, reputation, networks, or institutions. Persistent inequality would then require a different interpretation from initial inequality.

Suppose institutions substantially attenuate such contingencies. The same research would point toward mobility and resilience rather than persistence.

And suppose strategy–outcome relationships transport poorly across time and situated positions. Success stories, benchmarking, management fashions, and policy imitation would become weaker sources of general causal knowledge.

These alternatives matter because societies routinely convert outcomes into judgments about merit, competence, deservingness, and expertise.

Different empirical worlds justify different social conclusions. Those empirical dependencies should be made explicit before judgments about merit, expertise, or institutions are drawn.

12. What Can Research on Luck Realistically Accomplish?

Research on luck may never predict who will get lucky next. Prediction may be the wrong ambition.

It can still do useful work.

It can distinguish a good strategy from a good realized outcome. It can separate an initial divergence from the mechanisms that later amplify it. It can study chance events in careers, serendipity in discovery, statistical

variation in investment performance, and cumulative advantage in social systems. It can test whether successful strategies transport across populations and periods. It can identify cases in which positive evidence supports a particular realization mechanism—and cases in which attribution remains underidentified.

The sequence developed in this essay is deliberately simple:

strategy → possible states → realization → outcome → attribution → social learning → feedback.

A realized outcome is not the same thing as the strategy that preceded it.

An unexplained outcome is not evidence of luck.

A rare outcome is not a definition of luck.

Repeated success can reveal capability while also changing the conditions of future success.

A successful history can contain useful knowledge without automatically becoming transportable advice.

And an explanation of the past can change the future when other people begin acting on it.

A formula assigning precise percentages of success to skill and fortune may be beyond what the evidence can support.

What remains possible is attributional discipline: separating what people chose, what could have happened, what actually happened, what followed from it, and what we later decided the story meant.

In a world that routinely turns winners into experts, outcomes into merit, and histories into prescriptions, that discipline matters.

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